

Michael Newman

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Projects

Full Stack Project: SJSU - BQ ([blog](#))

- Created a full stack discussion forum application using **Java**, **Apache Tomcat**, **HTML**, and **MySQL** with a team of engineers enabling users to ask and answer posted questions seamlessly.
- Architected the system with **Apache Tomcat** and **MySQL**, implementing a live **relational database** for efficient management of questions, user data, and permissions
- Integrated **PHP** scripts to interface with the **MySQL** database, facilitating smooth database queries and secure login functionality

3D Action RPG: Fateful Fragments ([blog](#) - password: 'mnew')

- Developed a deliverable prototype of a **3D** action RPG in the **Unity Game Engine** using **C#**, featuring fast-paced hack-and-slash soul's-like combat
- Programmed my own **Finite State Machines** in **C#** to create custom dynamic **AI agent** behaviors
- Utilized Unity's **layer masks** and **finite state machine animator controllers** for seamless animation blending through code
- Created four immersive **boss battles** with multiple phases, attack types, and particle systems

Custom Game Engine: 3D Moon Lander Simulator ([blog](#))

- Engineered my own **3D game engine** and moon landing simulation game using **C++** and **openFrameworks**, with advanced physics, collision detection, cinematic lighting, and particle systems
- Leveraged complex **vector and matrix calculations** and an **octree** data structure to ensure precise physics simulation and collision accuracy

Education

San Jose State University | B.S. Computer Science

August 2019 - August 2022

Skills

Languages	Tools
• Java • C# • C++ • MySQL • Javascript	• Microsoft Visual Studio Code • Github • React • JSON, XML • Jira

Work Experience

Studio Mnew - Gameplay Programmer	July 2024 - Current
• Programmed and tuned core gameplay systems using Unity and C# such as the player controller, AI behavior, progression systems, and action combat • Collaborated with my team to establish and document best code practices and delivery of assets	
Costco Wholesale - Sales Assistant	June 2023 - July 2024
• Propelled sales by assisting customers with inquiries about televisions, laptops, jewelry, and other products • Stayed abreast of the latest technology trends, sharing insights and knowledge with members	